



AGROPTICS

Agroptics and WaPOR Integration for Monitoring Land and Water Productivity

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Meet the Expert!



AGROPTICS

Ahmed Hashem, Ph.D.

- The founder of Agroptics
- 25 years of experience in irrigation engineering and precision agriculture
- Senior irrigation engineer and Professor
- Purdue University, Suez Canal University
- Professional societies: MIAG, ASCE, USCID, CACD, and the ASABE
- Grew up on a farm, and was responsible for the irrigation ditches on our farm



Term project Team

Suez Canal University, Agricultural Engineering



Fatma Fekry



Mohamed Khaled



Mohamed Osama



Hassan Abo-Elnassar



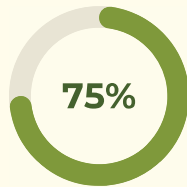
Nourhan Mohamed



Ziad Ali

The Problem: Farmers & Water Organizations do not have the tools

Traditional Decision Making



Decisions are based on visual conditions or a calendar method

Technology Gap



Technology advancement and technology adoption

Annual Net Loss



Water waste, pumping costs, labor, and agrochemical losses

Agroptics offers cost-effective solution to efficient water management

Organization scale



Customized Solutions

**WaPOR
Integration**



**Golf
Courses**

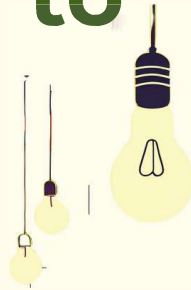


Others....

Water quality
Canal tracing
etc....

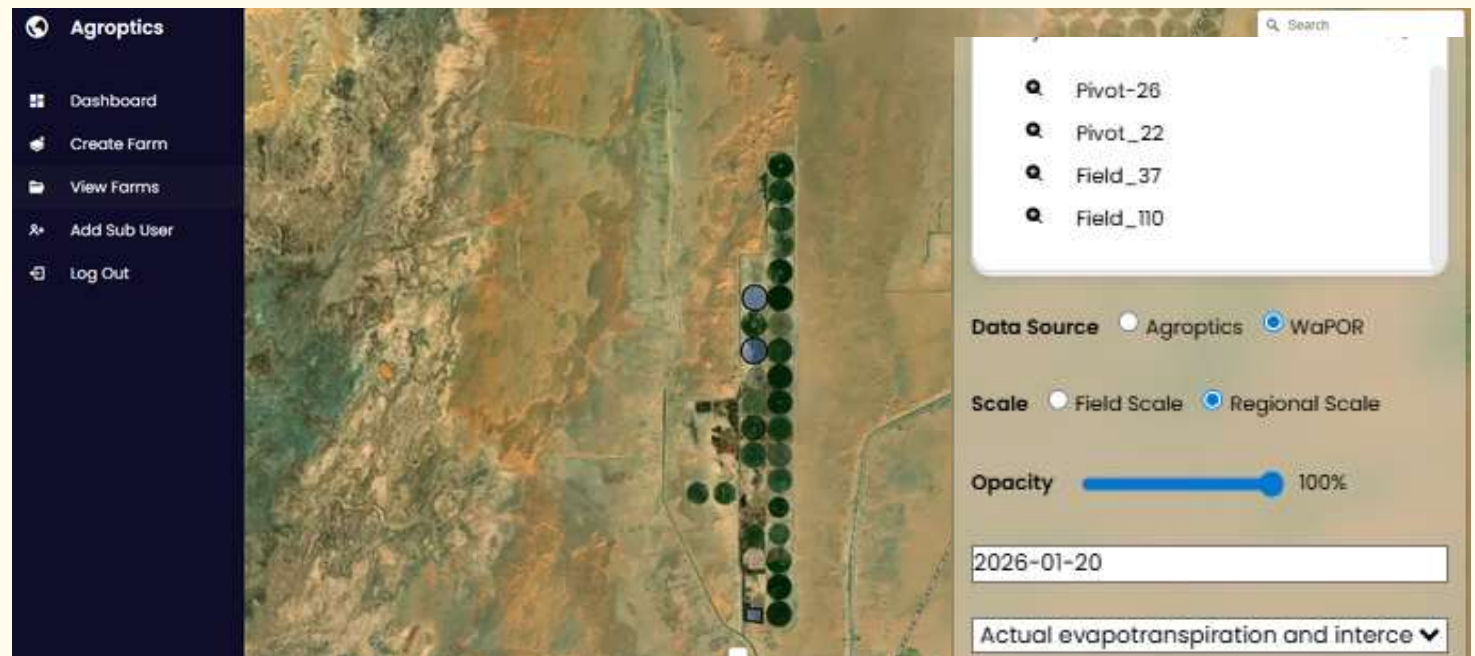


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Organization scale

WaPOR Integration



Agroptics offers cost-effective solution to efficient water management

Farm scale



**Real-time, data-driven
irrigation decision-making**



**Forecast consumptive
water use**

7-day irrigation schedule forecast



**Cumulative consumptive
water use**



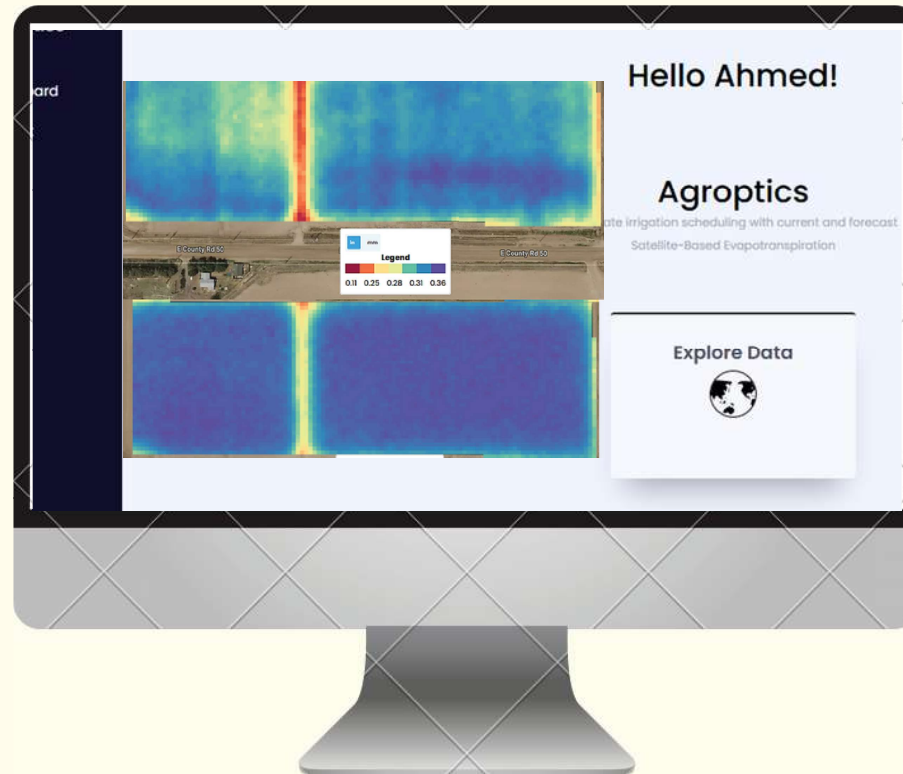
Does not require sensors



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Our Services-Farm Scale

Irrigation Management using remote sensing



Consultation



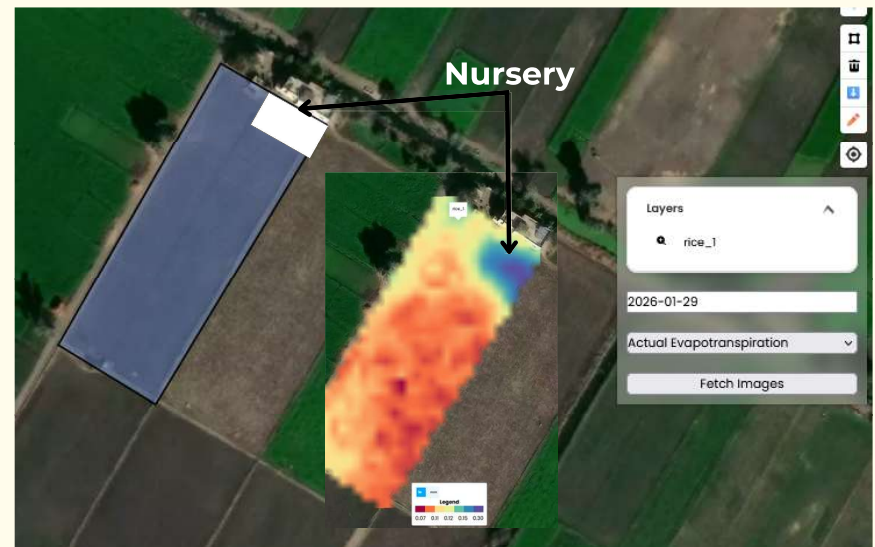
**Irrigation systems
infrastructure
improvements
40% increase in yield**

Field Name	Actual Evapotranspiration (in)	Next Irrigation Event	Next Irrigation Amount (in)
Field_24-Alfalfa	0.3	2025-06-13	3
Field_17_onion	0.3	2025-06-14	1.6
Field_15_Onio	0.1	2025-06-14	1.8
Field_22_Alfalfa	0.3	2025-06-16	3
Field_19_Onion	0.2	2025-06-16	1.6
Field_18_Onion	0.2	2025-06-16	1.6
Field_23_alfalfa	0.1	2025-06-16	3
Field_9_alfalfa	0.2	None within the next 7 days	None within the next 7 days

Case Study #1

Comparison of measured water diversion, WaPOR, and Agroptics

- Measured applied irrigation water for rice field, Research farm, Water Management Research Institute, Zankalon, Al-Sharqia.
- Rice field, 1.2-acre ($\sim 4856 \text{ m}^2$)
- Growing Season: May 31, 2025, October 31, 2025
- Nursery $\sim 300 \text{ m}^2$
- Clay soil
- Surface irrigation (basin),
- Agroptics Vs. WaPOR ET,
- Agroptics and actual applied irrigation & schedule were compared



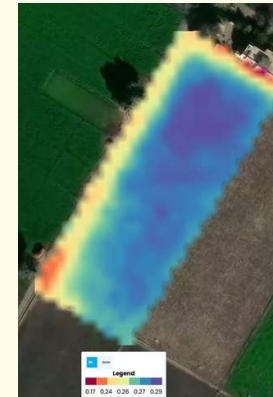
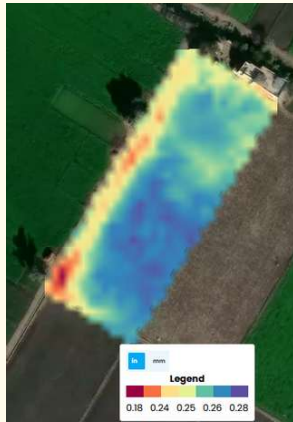
Case Study # 1

Comparison of measured water diversion, WaPOR, and Agroptrics

July 31

August 30

•ET



•Dekadal ETI

(July 20-31)



(Aug 20-31)



Case Study #1

Comparison of measured water diversion, WaPOR, and Agroptrics

July 31

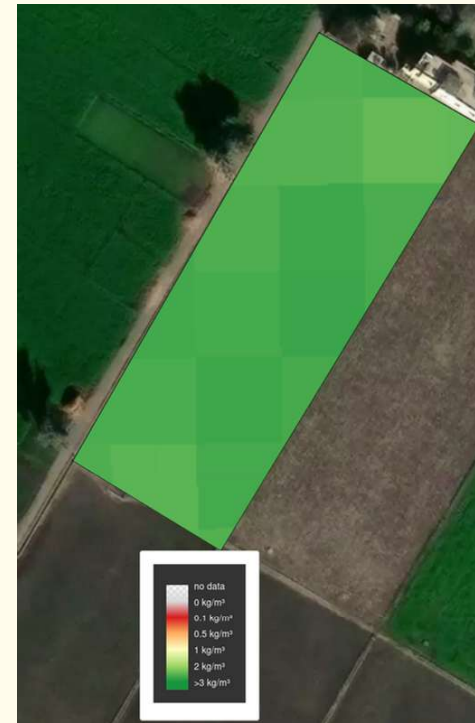
•Net Biomass water productivity

(July 20-31)



August

(Aug 20-31)



Case Study # 1

Comparison of measured water diversion, WaPOR, and Agroptrics

- **Agroptrics VS. WaPOR ETc comparison**
 - Cumulative
 - Dekadal time series
- **Applied Water (Agroptrics Vs. Actual)**
- **Total number of recommended irrigation Vs actual irrigations**
- **Recommended irrigation Vs actual irrigation timing**



Case Study # 1

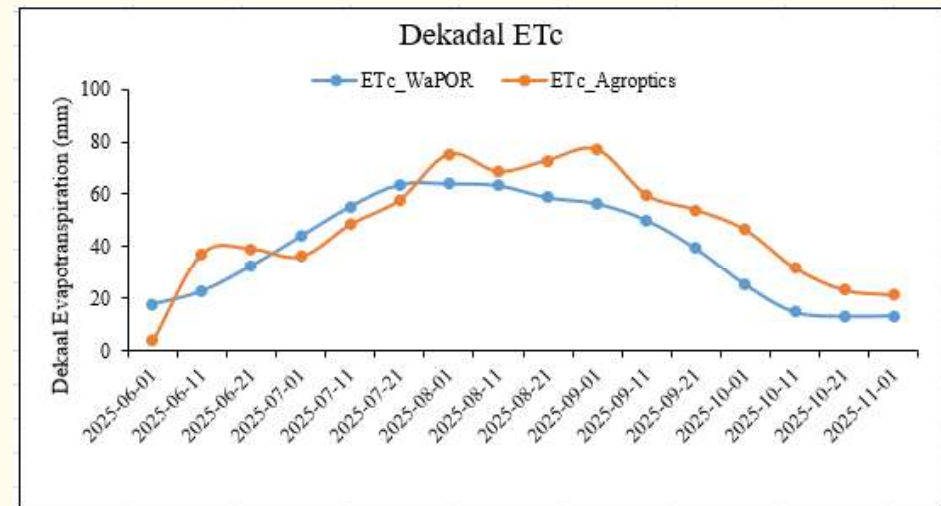
Comparison of measured water diversion, WaPOR, and Agroptics

Agroptics VS. WaPOR ETc comparison

Cumulative



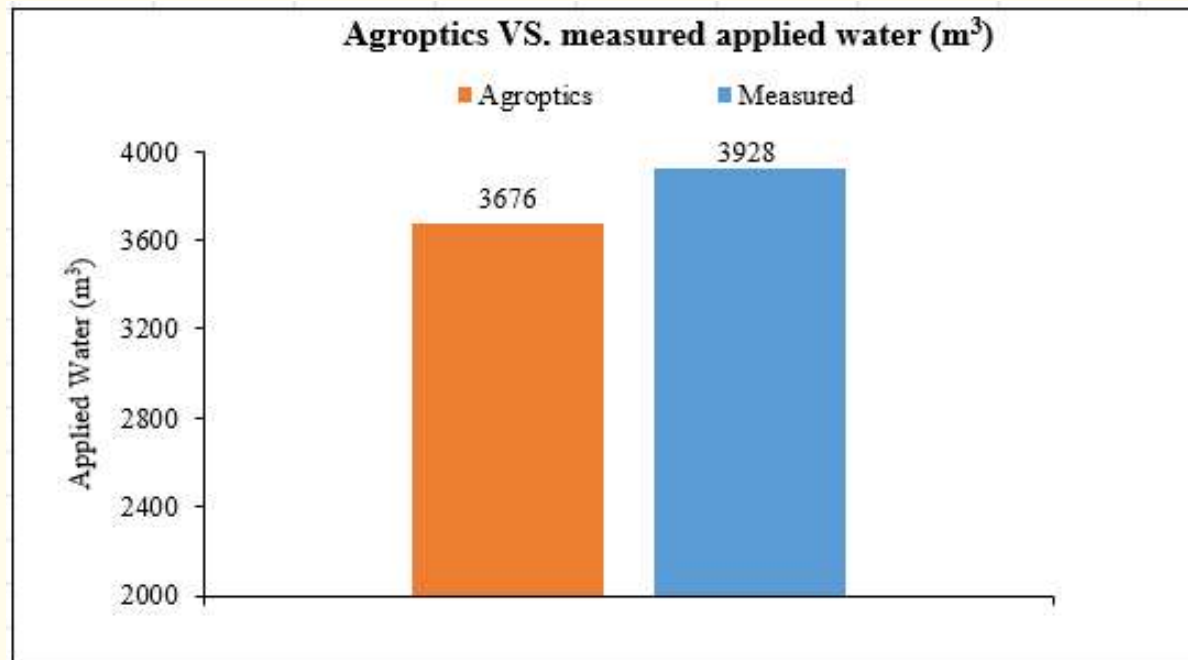
Dekadal time series



Case Study # 1

Comparison of measured water diversion, WaPOR, and Agroptics

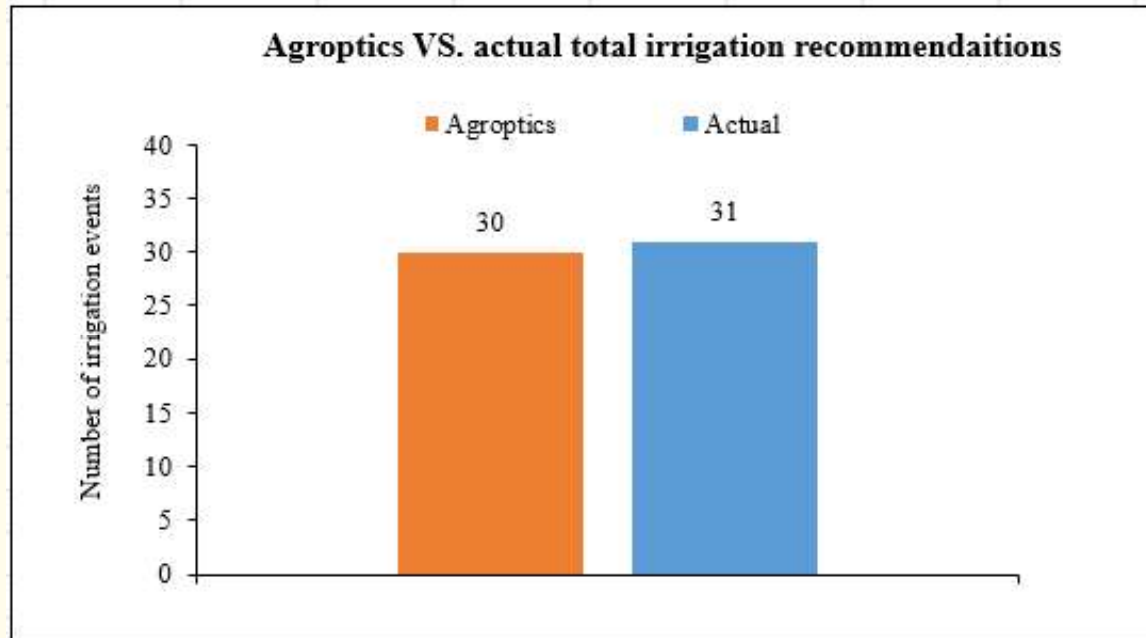
Applied Water (Agroptics Vs. Actual)



Case Study # 1

Comparison of measured water diversion, WaPOR, and Agroptics

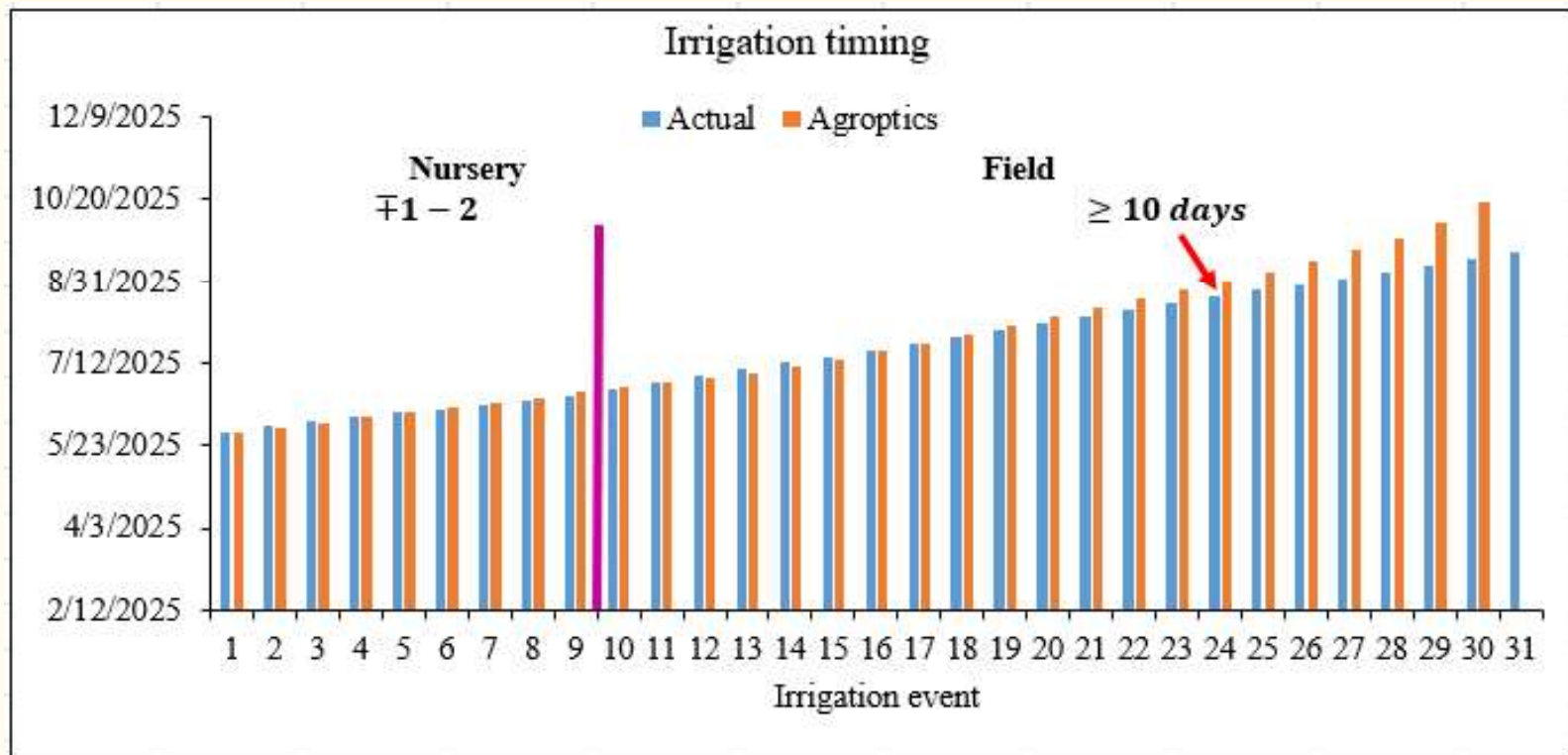
Total number of Agroptics recommended irrigation Vs actual irrigations



Case Study # 1

Comparison of measured water diversion, WaPOR, and Agroptics

Recommended irrigation Vs actual irrigation timing



Case Study # 1

- **In Summary**

- **WaPOR ETc and Biomas are valuable tools for irrigation management under Egyptian conditions**
- **The combination of Agroptics, WaPOR, and the measured data is valuable for field-scale accuracy assessment studies**
- **Agroptics and applied irrigation events are approximately the same**
- **Irrigation timing needs more calibration and validation, especially toward the end of the season**
- **Further water productivity assessment with the measured data will be performed**
- **Term project students did an excellent job on the WaPOR data download and interpretation**



Next Steps

- **WaPOR irrigation schedule assessment**
- **Compare the measured yield and biomass with WaPOR data**
- **More calibration and validation for irrigation timing, especially towards the end of the growing season**
- **Additional research farms' data (NWRC), with various crops and geographic locations, will be assessed VS. Agroptics**
- **Looking for collaboration opportunities**
- **Others.....**

Key Metrics

Proven economic impact & resource savings

On Average, savings in 2 years (~ 3,000 acres to date)



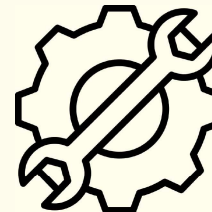
~ **5.7 million** m^3 of
water savings
\$276 K



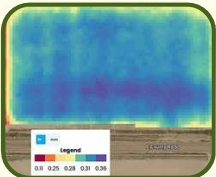
Less Pumping hours
(**> 10,000 hours; energy bills**)
~\$9-45 K (pump capacity)



Less labor hours
(**> 2,000 hours**)
~\$40 K



~ **20% less**
maintenance



Irrigation infrastructure improvements on 35 fields



Conclusion

Agroptics empowers growers, organizations, and agencies to take full control of their farming operations, helping optimize irrigation, reduce costs, and promote sustainable agriculture.



Acknowledgements

- The Food and Agriculture Organization of the United Nations (FAO) and Wageningen University (WU)
- The National Water Research Center (NWRC) and the Water Management Research Institute (WMRI)
- Dr. Mohamed Embaby and Dr. Magda Salah, for sharing the measured field data
- Dr. Hesham Aboelsoud for the WaPOR training
- Term project students
- Agroptics leadership for facilitating the data analysis and funding this study



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Let's schedule a call and talk!



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